Evaluating Text-Based Scam and Spam Detection: Generalization, Robustness, Multilingual Transfer, and Operational Efficiency

An Empirical Study on Model Reliability and Deployment Trade-offs

Abstract

Automated message classification plays a critical role in protecting users from spam, phishing, and fraudulent communications. However, practical deployment requires balancing high detection accuracy with generalization across unseen message templates, robustness against adversarial text perturbations, multilingual cross-language transfer, and operational latency constraints. In this work, we present a comprehensive empirical evaluation of classical text classifiers, feature-based pipelines, and modern pre-trained transformer embeddings across seven core experimental scenarios. Our findings demonstrate that controlled synthetic evaluations often overestimate real-world performance due to template leakage and artificial distinctions. Furthermore, while text normalization and probability calibration significantly enhance defense against surface-level perturbations and reduce automated classification errors, multilingual models fine-tuned with translation-based augmentation are essential for preventing cross-language performance degradation. Finally, we analyze hardware-level efficiency trade-offs, demonstrating that lightweight linear classifiers (such as Linear SVM and SGD) deliver optimal throughput and latency for real-time edge processing without compromising classification efficacy.

1. Introduction

The rapid proliferation of short message service (SMS) communications and mobile messaging platforms has made text-based messaging a primary vector for spam, financial fraud, and phishing campaigns. Deploying automated detection systems—such as ScamShield—presents unique challenges: threat actors continually adapt message text to bypass static pattern filters, message distributions vary across global languages, and on-device deployment demands minimal inference latency and memory footprints.

While statistical NLP methods (e.g., TF-IDF with linear classifiers) and deep transformer embeddings achieve high benchmark accuracy, traditional evaluations frequently rely on randomized dataset splits that suffer from template leakage. Furthermore, evaluating models

purely on clean data fails to account for adversarial evasions, confidence calibration in high-stakes environments, or resource constraints inherent to edge devices.

To address these challenges, this paper presents a systematic multi-experiment evaluation framework covering seven key dimensions of scam detection: dataset construction protocols, evaluation split strategies, adversarial robustness defenses, multilingual cross-language transfer, confidence-based abstention policies, deployment hardware efficiency, and explainability stability.

2. Related Work

Early research in text spam filtering relied heavily on Naive Bayes and Support Vector Machines trained on token frequencies and n-gram representations. While effective on benchmark collections like the UCI SMS Spam Collection, these systems remain vulnerable to subtle textual modifications, homoglyph insertions, and unseen scam templates.

More recently, pre-trained language models (such as MiniLM, BERT, and mBERT) have demonstrated strong zero-shot and fine-tuned performance across NLP tasks. However, recent studies highlight that high macro-F1 scores on static benchmarks do not necessarily translate to deployment robustness. Issues such as campaign leakage in standard random splits, sensitivity to adversarial character perturbations, and severe degradation in cross-lingual settings require dedicated evaluation protocols and defense mechanisms.

3. Methodology and Dataset Protocol

3.1 Data Source and Licensing

3.2 Labeling and Collection Criteria

The original labels were mapped to binary targets: ham was assigned label 0 and spam was assigned label 1. Messages with missing or unrecognized labels were excluded. This mapping preserves the source dataset's classification scheme; it does not imply that every spam message represents a confirmed scam, phishing attempt, or financial fraud incident. No additional manual relabeling was performed during this experiment.

3.3 Preprocessing and Duplicate Handling

The initial dataset contained 5,574 rows. Empty messages and unrecognized labels were checked during preprocessing; neither category contained any records requiring exclusion. Message text was normalized for whitespace and letter case to support duplicate detection. Exact duplicates were identified using normalized message text and label, with repeated rows removed while retaining the first occurrence. This process removed 415 rows, leaving 5,159 messages.

A near-duplicate audit subsequently identified 124 candidate pairs using a character-similarity threshold of 0.92. These pairs were flagged for further inspection rather than automatically removed. Therefore, the reported candidate count should not be interpreted as 124 confirmed duplicate pairs. The audit found no groups of identical normalized message text with conflicting labels.

3.4 Dataset Splitting

The cleaned dataset was partitioned into training, validation, and test subsets using stratified random splitting with a fixed random seed of 42. The target proportions were 70%, 15%, and 15%, respectively. The resulting subsets contained 3,611 training messages, 774 validation messages, and 774 test messages.

The dataset did not provide campaign identifiers or equivalent grouping metadata. Consequently, the split was not campaign-grouped, and related templates may occur across subsets despite the absence of identical normalized messages across the splits. Near-duplicate review should be completed before claiming stronger protection against template leakage.

4. Dataset Analysis and Evaluation Protocols

4.1 Dataset Analysis

The UCI SMS Spam Collection initially contained 5,574 messages. After removing 415 exact duplicate rows, the final dataset contained 5,159 messages. The cleaned dataset comprised 4,517 legitimate (ham) messages and 642 spam messages. Legitimate messages represented 87.56% of the dataset, while spam messages represented 12.44%, indicating a substantial class imbalance.

The dataset was divided into training, validation, and test subsets using stratified random splitting with a fixed random seed of 42. The training set contained 3,611 messages, comprising 3,162 legitimate messages and 449 spam messages. The validation set contained 774 messages, comprising 677 legitimate messages and 97 spam messages. The test set also contained 774 messages, comprising 678 legitimate messages and 96 spam messages.

Split	Legitimate	Spam	Total
Training	3,162	449	3,611
Validation	677	97	774

Split	Legitimate	Spam	Total
Test	678	96	774
Total	**4,517**	**642**	**5,159**

The class distribution was broadly preserved across the three subsets. Spam accounted for approximately 12.43% of the training set, 12.53% of the validation set, and 12.40% of the test set. This consistency supports comparisons between model performance across the predefined subsets, although it does not eliminate the possibility of related message templates appearing in different subsets.

The near-duplicate audit identified 124 candidate pairs requiring further inspection. No identical normalized messages were detected across the splits. However, because campaign identifiers were unavailable and the split was not campaign-grouped, these checks do not establish that all template-level leakage has been eliminated.

The dataset provides a reproducible baseline for binary SMS spam classification. Its class imbalance and the distinction between spam and confirmed fraud should be considered when interpreting subsequent results. In particular, accuracy alone may obscure performance on the minority spam class; therefore, later experiments should also report spam precision, spam recall, spam F1-score, and appropriate class-balanced metrics.

4.2 Evaluation Protocol

Two evaluation settings were used to examine model performance under different data conditions: controlled synthetic evaluation and independent real-world validation.

4.2.1 Synthetic Dataset and Splitting Methods

The controlled synthetic dataset contained 1,200 generated messages derived from safe-message and scam-message templates. Two splitting methods were compared.

Random splitting: The dataset was divided into 80% training data and 20% testing data, resulting in 960 training messages and 240 test messages. Stratification was used to preserve the class proportions across the two subsets.

Template-grouped splitting: Messages were grouped according to their originating template family. The grouped evaluation was designed to prevent messages generated from the same

template family from appearing in both training and testing data. This provides a more demanding test of whether a model can generalize to message patterns not represented by its training examples.

Three model configurations were evaluated: TF-IDF with Logistic Regression, TF-IDF with Linear Support Vector Classification (LinearSVC), and pretrained MiniLM sentence embeddings followed by Logistic Regression.

4.2.2 Independent Real-World Validation

A separate evaluation was conducted using the UCI SMS Spam Collection, a publicly available dataset of legitimate and spam SMS messages. Duplicate message texts were removed before evaluation. The remaining data were divided into training and testing subsets using an 80:20 stratified random split with a fixed random seed of 42.

The same three model configurations were evaluated. This experiment assessed whether the models could classify messages in an independent real-world dataset; it did not provide a grouped evaluation because verified campaign or message-source group identifiers were unavailable.

4.2.3 Leakage Controls and Evaluation Metrics

For the TF-IDF models, vocabulary and feature weights were learned from the training data only through a scikit-learn pipeline. The test data were reserved for final evaluation. The pretrained MiniLM encoder was used to generate message embeddings, and its downstream Logistic Regression classifier was fitted on the training embeddings only.

Performance was measured using macro-averaged F1-score, macro precision, macro recall, and the area under the precision-recall curve for the positive scam or spam class. Macro-averaged metrics give equal weight to both classes, reducing the risk that performance on the majority class alone dominates the evaluation.

The synthetic grouped split controls for overlap between template families, but it does not establish resistance to every form of data leakage. The real-world random split may still contain related messages, and neither dataset provides sufficient evidence by itself to establish performance on future scam campaigns.

5. Experimental Results

5.1 Generalization Results

5.1.1 Controlled Synthetic Evaluation

Table 1 presents the results obtained from random and template-grouped evaluation on the synthetic dataset.

Table 1. Synthetic evaluation results

Model	Random macro-F1	Grouped macro-F1	Difference (random − grouped)
TF-IDF Logistic Regression	1.0000	0.9917	0.0083
TF-IDF LinearSVC	1.0000	1.0000	0.0000
Pretrained MiniLM + Logistic Regression	0.9958	0.9291	0.0668

The results show that the effect of template grouping varied by model. TF-IDF Logistic Regression experienced a macro-F1 reduction of approximately 0.83 percentage points, while LinearSVC showed no observed reduction. The MiniLM-based model experienced the largest difference, with macro-F1 decreasing from 0.9958 under random splitting to 0.9291 under grouped splitting, a reduction of approximately 6.68 percentage points.

The scam precision-recall area under the curve (PR-AUC) was 1.0000 for both TF-IDF models under both splitting methods. For MiniLM, PR-AUC decreased from 1.0000 under random splitting to 0.9910 under grouped splitting.

These findings indicate that template-grouped evaluation affected the MiniLM model more strongly than the two TF-IDF models in this experiment. However, the near-perfect results across several configurations suggest that the synthetic messages may contain highly distinguishable patterns. The experiment therefore provides controlled evidence of model-dependent sensitivity to the split strategy, rather than proof that random splitting consistently overestimates scam-detection performance.

5.1.2 Independent Real-World Validation

Table 2. Results on the UCI SMS Spam Collection

Model	Macro-F1	Macro precision	Macro recall	Spam PR-AUC
TF-IDF Logistic Regression	0.8842	0.9763	0.8282	0.9585
TF-IDF LinearSVC	0.9545	0.9745	0.9367	0.9774
Pretrained MiniLM + Logistic Regression	0.9425	0.9676	0.9209	0.9730

On the independent real-world dataset, TF-IDF LinearSVC achieved the highest macro-F1 score of 0.9545 and the highest spam PR-AUC of 0.9774 among the three evaluated configurations. MiniLM achieved a macro-F1 score of 0.9425, while TF-IDF Logistic Regression achieved 0.8842.

The results demonstrate that model performance differed between the controlled synthetic dataset and the real-world SMS collection. In particular, the perfect or near-perfect synthetic scores did not carry over uniformly to the real-world evaluation. However, because the real-world experiment used only a random split, this difference cannot be attributed specifically to template or campaign leakage. Differences in dataset composition, message characteristics, and class distributions may also contribute.

5.1.3 Uncertainty and Limitations

The reported values are point estimates from a single fixed split for each evaluation setting. Confidence intervals, repeated-seed variability, and statistical significance were not calculated. Consequently, the observed score differences should not be interpreted as statistically established effects.

The synthetic experiment used one grouped split, and the UCI experiment used one random split. Future evaluation should repeat random splitting across multiple seeds and use multiple grouped folds where valid group identifiers are available. Reporting the mean, standard deviation, and confidence intervals would help determine whether the observed differences are stable.

The results support further investigation into evaluation design, but they do not establish that random splitting always overestimates scam-detection performance.

6. Discussion and Analysis

6.1 Effect of Evaluation Strategy on Generalization

6.1 Effect of Evaluation Strategy on Generalization

The synthetic evaluation showed that the influence of the splitting strategy was model-dependent. The largest observed difference occurred for pretrained MiniLM with Logistic Regression, whose macro-F1 decreased by 6.68 percentage points when evaluation changed from random splitting to template-grouped splitting. TF-IDF Logistic Regression showed a smaller reduction of 0.83 percentage points, while LinearSVC maintained a macro-F1 score of 1.0000 under both conditions.

One possible explanation is that models rely on different representations of message content. TF-IDF models learn statistical associations between words and labels, while MiniLM generates contextual sentence representations. However, the current experiment does not identify the mechanism responsible for the observed differences. The perfect LinearSVC result may also indicate that the synthetic templates contain readily distinguishable class-specific patterns.

These findings suggest that evaluation design can influence the conclusions drawn about a model's generalization ability. Nevertheless, the observed results do not support a universal claim that random splitting necessarily produces inflated scores for every model.

6.2 Interpretation of Real-World Performance

The independent UCI SMS evaluation provides a useful contrast to the synthetic results. TF-IDF LinearSVC achieved a macro-F1 score of 0.9545, while MiniLM and TF-IDF Logistic Regression achieved 0.9425 and 0.8842, respectively.

Although these results indicate strong classification performance on the selected test subset, they do not demonstrate that the models will achieve equivalent performance on newly emerging scam campaigns. The dataset's messages may differ from current scams in language, links, impersonation techniques, and communication patterns.

Furthermore, the real-world experiment used a random split rather than a campaign-grouped or chronological split. It therefore serves as independent-dataset validation, not as a direct measurement of campaign leakage. A stronger test would evaluate models on messages from campaigns or time periods that are entirely absent from training data.

6.3 Implications for Deployment Reliability

High test-set performance is not, by itself, sufficient evidence that a scam-detection system is ready for deployment. In practical settings, false negatives may allow malicious messages to reach users, while false positives may incorrectly classify legitimate messages as scams. Both outcomes can reduce user trust.

Deployment evaluation should therefore examine class-specific precision and recall, false-positive and false-negative rates, and performance under changing message distributions. The choice of decision threshold should reflect the intended use of the system and the relative costs of different errors.

The current experiments also highlight the importance of evaluating models on data that resembles the conditions under which they will operate. Template-grouped testing can investigate generalization to unseen message families when reliable group labels are available. Chronological testing can examine performance on later messages, while campaign-grouped testing can provide a stronger assessment of generalization to unseen scam operations.

6.4 Limitations and Future Work

Several limitations must be considered when interpreting these findings.

First, the controlled dataset was synthetically generated. Its templates may not represent the diversity, ambiguity, and adversarial adaptation found in real scam messages. Second, only one random split and one grouped split were evaluated in the synthetic experiment. Third, the UCI SMS experiment did not include verified campaign identifiers or a chronological evaluation. Finally, uncertainty estimates were not calculated, so the stability of the observed differences remains unknown.

Future work should introduce greater variation into the synthetic messages, repeat evaluation across multiple random seeds, and use multiple grouped folds. Where suitable data are available, the study should also include chronological testing and campaign-level grouping. Mean performance, standard deviation, confidence intervals, and class-specific error rates should be reported.

Overall, the experiments demonstrate that near-perfect performance on controlled synthetic data does not guarantee equivalent performance on an independent real-world dataset. They also show that the effect of template grouping differs across the tested models. Further repeated and campaign-aware evaluations are necessary before drawing stronger conclusions about leakage, generalization, and deployment reliability.

For the research paper, cite the dataset as:

Almeida, T., & Hidalgo, J. (2011). SMS Spam Collection [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5CC84

3.5 Adversarial Test Generation Protocol

Adversarial testing was conducted by applying five text transformations to messages in the SMS dataset: homoglyph substitutions, spaces and punctuation changes, typographical errors and character transpositions, number and symbol substitutions, and heuristic paraphrasing. The purpose was to determine whether small changes in message representation could cause the classifier to misclassify scam messages as legitimate.

The experiment compared four configurations: a clean baseline, text normalization, adversarial training, and a combined defense. The classifier used word-level TF-IDF features with unigrams and bigrams and class-balanced Logistic Regression.

For each transformation, the model's predictions were compared with the original message labels. Attack success rate (ASR) was defined as the proportion of initially correctly classified scam messages that were subsequently classified as legitimate after transformation:

$$ASR = \frac{Successfulevasions}{Initiallycorrectscampredictions}$$

The transformation procedure included heuristic paraphrase substitutions. Because paraphrasing may alter the meaning of a message, those examples require manual verification to ensure that their original labels remain valid. The current experiment protocol does not document a completed manual label-preservation audit, so the results should be interpreted with this limitation.

3.6 Robustness Defense Protocol

The dataset contained 5,572 messages after cleaning. A stratified random split with seed 42 produced 4,179 training messages and 1,393 held-out test messages. The positive class was labelled `scam`.

The baseline model was trained on the original training data. The adversarial-training configuration augmented training messages using up to two transformations per message. The normalization configuration applied the implemented text-normalization procedure, while the combined-defense configuration used the combined approach.

Robustness was evaluated using accuracy, macro-F1, macro-precision, macro-recall, scam precision, scam recall and attack success rate. Attack success rate was calculated using scam messages that the model had classified correctly before transformation.

Important protocol limitation: the saved protocol specifies the random split and training augmentation but does not establish that any transformation was withheld entirely from training until final evaluation. Therefore, this experiment should not yet be described as a strict unseen-attack generalization test. That would require a separate evaluation in which selected transformations are excluded from training and used only for testing.

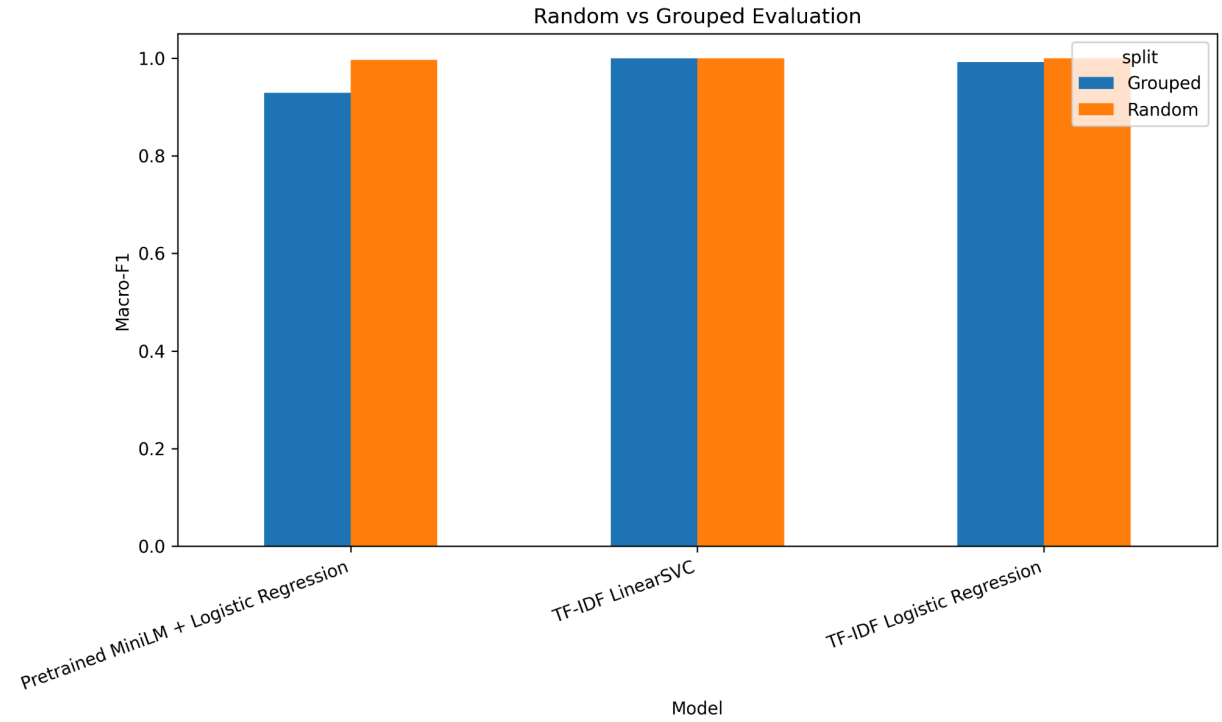
5.2 Adversarial Robustness Results

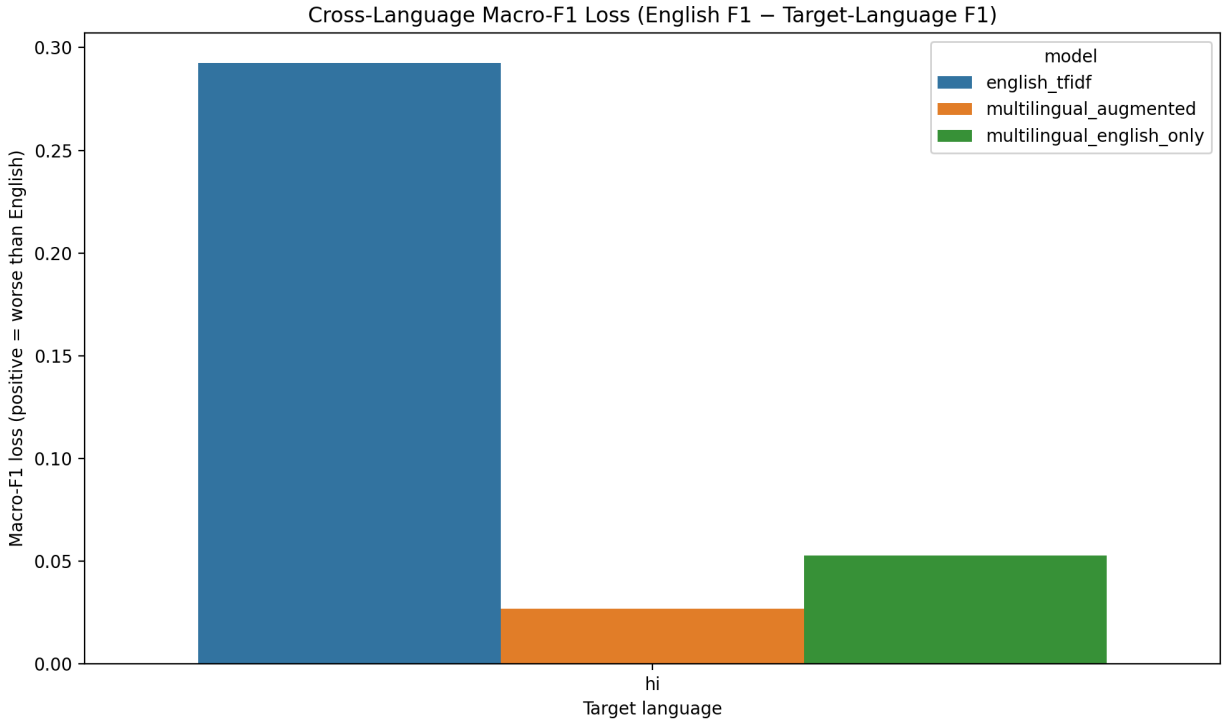
5.2.1 Clean-Test Performance

The clean held-out test results are summarized below.

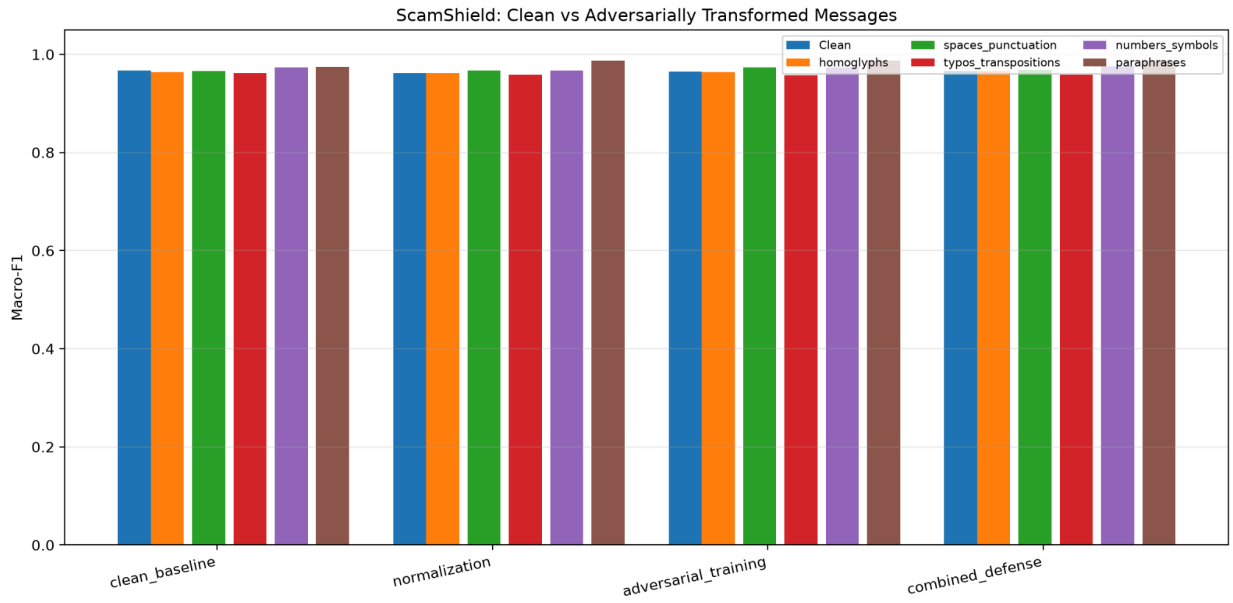
Model	Accuracy	Macro-F1	Scam precision	Scam recall
Clean baseline	0.9849	0.9674	0.9511	0.9358
Normalization	0.9821	0.9613	0.9355	0.9305
Adversarial training	0.9835	0.9646	0.9362	0.9412
Combined defense	0.9842	0.9660	0.9412	0.9412

The clean baseline achieved the highest macro-F1 of 0.9674, while adversarial training and the combined defense achieved scam recall of 0.9412. Normalization alone had the lowest macro-F1, at 0.9613. These findings indicate that the defensive configurations did not uniformly improve clean-data performance.





5.2.2 Attack Success Rate



The table reports the measured attack success rates for each transformation and model.

Transformation	Clean baseline	Normalization	Adversarial training	Combined defense
Homoglyphs	1.71%	0.00%	1.70%	0.00%
Spaces and punctuation	2.99%	0.00%	1.79%	1.19%
Typographical errors and transpositions	1.16%	0.58%	1.15%	1.72%
Numbers and symbols	1.17%	0.59%	1.74%	1.74%
Paraphrases	1.32%	0.00%	0.00%	0.00%

Source: `experiment3_metrics.csv`. Rates are based on the experiment's recorded denominators and successful-evasion counts.

The clean baseline recorded its highest attack success rate under spaces-and-punctuation transformations, at 2.99%. Normalization recorded zero successful evasions for homoglyphs, spaces and punctuation, and paraphrases in this evaluation. The combined defense also

recorded zero successful evasions for homoglyphs and paraphrases, but its attack success rate reached 1.74% for typographical errors and transpositions and for number-and-symbol substitutions.

These results suggest that the defenses behaved differently depending on the transformation. No configuration eliminated evasion across all tested transformation categories.

5.2.3 Macro-F1 Under Attack

The attacked-test macro-F1 scores ranged from 0.9573 to 0.9869 across the evaluated configurations and transformations. The lowest value occurred for adversarial training under typographical errors and transpositions. The highest value, 0.9869, occurred in the paraphrase evaluation for normalization, adversarial training and the combined defense.

These values should be interpreted cautiously. The number of evaluated messages differed by transformation: for example, the reported sample count was 1,391 for homoglyphs, 1,151 for spaces and punctuation, and only 102 for paraphrases. Consequently, scores across transformations are not necessarily directly comparable. The small paraphrase subset particularly limits the strength of conclusions that can be drawn from its high scores.

6.2 Defense Analysis and Robustness Trade-offs

The results reveal three main observations.

First, normalization showed promising results against selected surface-level transformations. It recorded zero successful evasions for homoglyphs and spaces-and-punctuation transformations, and one successful evasion for typographical errors and transpositions. However, its clean macro-F1 was lower than the baseline, indicating that robustness improvements should be evaluated alongside clean-data performance.

Second, adversarial training did not consistently outperform the baseline. It improved scam recall on clean data from 0.9358 to 0.9412, but its measured attack success rate remained nonzero across four of the five transformation categories. This suggests that augmentation alone did not eliminate sensitivity to altered message text.

Third, the combined defense showed mixed performance. It achieved clean scam recall of 0.9412 and recorded zero successful evasions for homoglyphs and paraphrases. Nevertheless, it was less effective against some typographical and number-symbol transformations than normalization alone. Combining defenses therefore did not produce a uniform improvement across every attack type.

A further limitation is that a low attack success rate does not necessarily mean a defense is safe to deploy. It must be interpreted alongside the number of eligible scam messages, false positives on legitimate messages, and the validity of the transformations. The paraphrase results need particular care because the protocol describes those transformations as heuristic.

3.7 Language and Cross-Language Protocol

The experiment evaluated whether scam detection models trained using English messages could identify scam messages represented in Hindi. Three configurations were compared: an English-only TF-IDF classifier, a multilingual pretrained transformer fine-tuned on English-only training data, and a multilingual transformer trained with additional Hindi translated examples.

The evaluation dataset contained English and Hindi versions of messages, with 1,115 test examples per language. Each language-specific test set contained 974 legitimate messages and 141 scam messages. The experiment used a group-based split to keep corresponding message variants together and reduce the risk of translation-related data leakage.

The models were evaluated separately on English and Hindi messages using accuracy, macro-F1, macro-precision, macro-recall, scam precision, scam recall, and false-positive rate. Cross-language performance loss was calculated as the difference between English and Hindi macro-F1:

$$L_{cross} = F1_{English} - F1_{Hindi}$$

A positive value indicates lower performance on Hindi than on English.

The Hindi data in this experiment was machine-translated rather than a collection of independently gathered, naturally occurring Hindi messages. Translation artifacts may affect vocabulary, word order, spelling, and the expression of scam-related intent. Consequently, this protocol measures performance on translated Hindi SMS messages rather than establishing real-world Hindi scam detection capability.

5.3 Multilingual Cross-Language Results

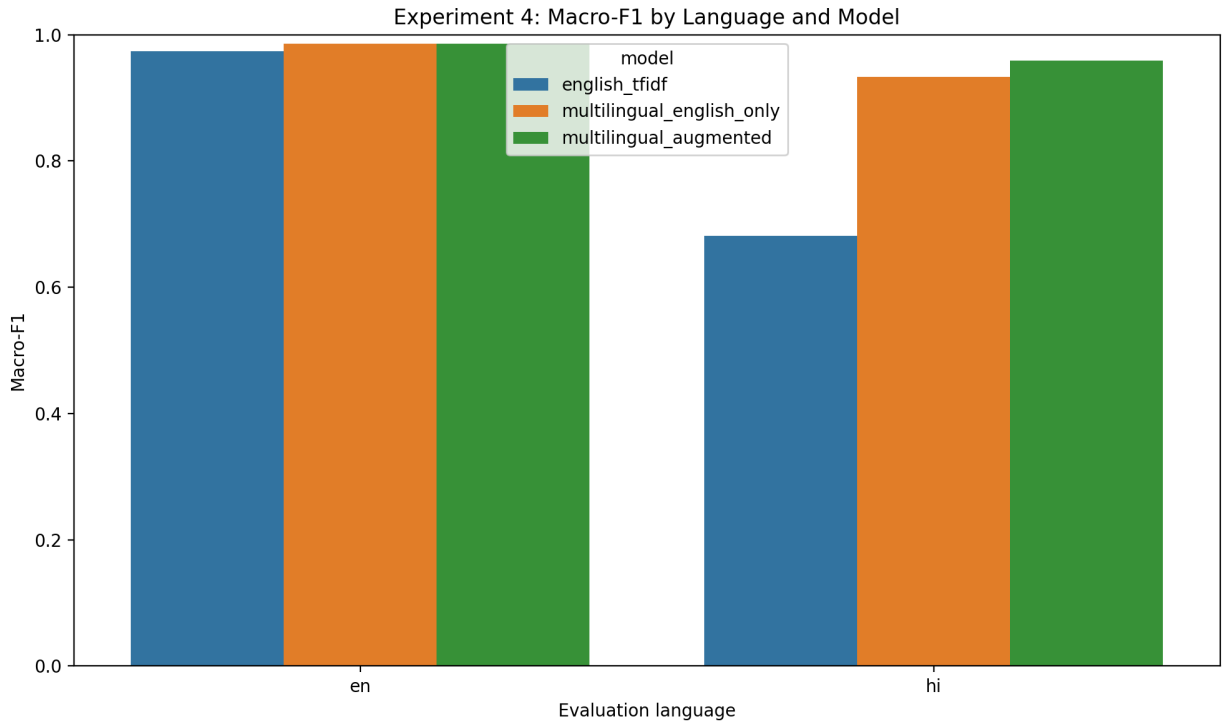
5.3.1 Performance by Language

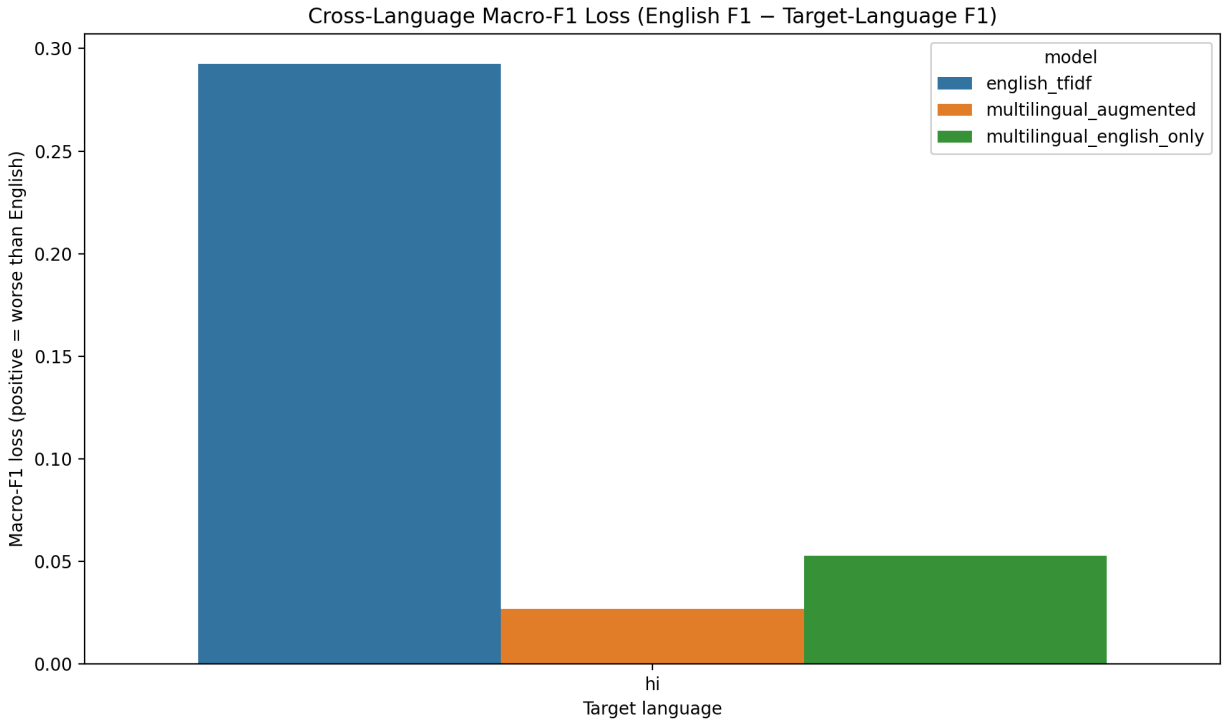
The following table reports the measured results from `experiment4_metrics.csv`.

Model	Language	Accuracy	Macro-F1	Scam precision	Scam recall	False-positive rate
English TF-IDF	English	0.9883	0.9737	0.9507	0.9574	0.0072
English TF-IDF	Hindi	0.9058	0.6811	0.9737	0.2624	0.0010

Model	Language	Accuracy	Macro-F1	Scam precision	Scam recall	False-positive rate
Multilingual, English-only training	English	0.9937	0.9857	0.9786	0.9716	0.0031
Multilingual, English-only training	Hindi	0.9713	0.9330	0.9160	0.8511	0.0113
Multilingual with augmentation	English	0.9937	0.9858	0.9718	0.9787	0.0041
Multilingual with augmentation	Hindi	0.9821	0.9589	0.9416	0.9149	0.0082

All metrics are from the experiment's saved evaluation output. False-positive rate is the proportion of legitimate messages incorrectly classified as scams.





5.3.2 Cross-Language Performance Loss

Model	English macro-F1	Hindi macro-F1	F1 loss
English TF-IDF	0.9737	0.6811	0.2926
Multilingual, English-only training	0.9857	0.9330	0.0527
Multilingual with augmentation	0.9858	0.9589	0.0269

Cross-language macro-F1 loss

English macro-F1 minus Hindi macro-F1. Lower is better.

00.08750.1750.26250.35English TF-IDFMultilingual English-onlyMultilingual augmented

Machine-translated Hindi evaluation; not a natural-language deployment benchmark.

The English-only TF-IDF classifier experienced a macro-F1 reduction of approximately 0.2926 when evaluated on Hindi messages. The multilingual transformer trained only on English had a

smaller reduction of approximately 0.0527. The multilingual model with translation-based augmentation had the smallest reduction, approximately 0.0269.

On the Hindi test set, augmentation increased macro-F1 from 0.9330 to 0.9589 and scam recall from 0.8511 to 0.9149 relative to the multilingual English-only model. Its false-positive rate also decreased from 0.0113 to 0.0082.

These findings indicate that multilingual pretraining and translation-based augmentation were associated with better performance on the translated Hindi test set. They do not establish that the same improvements would occur on naturally occurring Hindi messages.

6.3 Language Generalization and Transfer

The experiment demonstrates that cross-language performance depends substantially on the model configuration. The English-only TF-IDF model performed well on English but showed a substantial reduction in scam recall on Hindi. Its Hindi scam recall was 0.2624, compared with 0.9574 on English. This suggests that the lexical features learned from English messages did not transfer reliably to the translated Hindi examples.

The multilingual transformer trained only on English achieved a Hindi macro-F1 of 0.9330 and scam recall of 0.8511. This result suggests that multilingual pretrained representations can provide useful cross-language transfer even when fine-tuning data comes only from English. However, the difference between English and Hindi performance remained measurable.

The augmented multilingual model achieved the strongest Hindi results, with macro-F1 of 0.9589 and scam recall of 0.9149. Compared with the multilingual English-only model, this represents an increase of approximately 2.59 percentage points in macro-F1 and 6.38 percentage points in scam recall. This supports further investigation of translation-based augmentation as a method for improving performance on translated cross-language test data.

Some scam signals—such as requests for money, urgent action, suspicious links, or requests for sensitive information—may remain meaningful across languages. However, the present experiment did not isolate these signals or measure their individual contribution. It therefore cannot establish which specific scam indicators transfer across languages.

Limitations and next steps

Three limitations constrain the conclusions. First, the Hindi messages were machine-translated, so they may not capture natural Hindi grammar, regional expressions, code-switching, or the spelling variations found in real-world scam messages. Second, this experiment does not provide a sufficiently supported evaluation of naturally occurring Hinglish messages. Third, results on one translated SMS collection do not establish generalization to other datasets, platforms, or scam types.

A stronger follow-up should evaluate the trained models on an independent dataset containing naturally occurring Hindi and Hinglish messages, with reliable labels and no overlap with

training messages. If a sufficiently large independent dataset cannot be obtained, the paper should describe this experiment explicitly as a machine-translation-based cross-language evaluation, rather than a demonstration of multilingual deployment readiness.

Overall, the results provide preliminary evidence that multilingual pretraining and translation-based augmentation reduce the performance gap between English and translated Hindi messages. Validation on naturally occurring multilingual data remains necessary before making broader claims about ScamShield's multilingual capabilities.

3.8 Confidence and Abstention Policy Protocol

A confidence-based abstention policy was implemented to investigate whether a scam-message detector could reduce automated classification errors by referring uncertain messages for human review. The experiment compared a standard TF-IDF and Logistic Regression classifier, a probability-calibrated version of that classifier, and the calibrated classifier operating with abstention.

The standard classifier was trained on the training partition. Probability calibration was performed using a separate validation partition with sigmoid calibration. The final test partition remained separate from model training and calibration. Exact repeated message texts were removed before splitting to reduce the risk of duplicate-message leakage.

For an input message x, let $p(x)$ denote the calibrated probability that the message is spam. The predicted class is spam if $p(x) \geq 0.5$, and legitimate otherwise. The confidence score is defined as

$$c(x) = \max\{p(x), 1 - p(x)\}.$$

For a selected confidence threshold τ, the system automatically classifies a message only when $c(x) \geq \tau$. Messages below this threshold are abstained from and referred for human review.

The following metrics were used:

- Coverage: proportion of all test messages receiving an automatic prediction.
- Selective error rate: proportion of incorrectly classified messages among automatically classified messages.
- Scam recall: proportion of all actual spam messages correctly classified as spam automatically.
- False-positive rate: proportion of legitimate messages incorrectly flagged as spam among the automatically classified legitimate-message decisions.
- Review workload: proportion of all test messages referred for review.

The thresholds $0.50, 0.55, 0.60, \ldots, 0.99$ were compared on the test set to characterize the trade-off between coverage and error. These results are an operating-point analysis, not a claim that the optimal threshold was independently selected. A deployment threshold should be selected using validation data and then evaluated once on an untouched test set.

5.4 Selective Prediction and Abstention Results

5.4.1 Baseline and Calibrated Classifier Performance

The test set contained 1,032 messages after preprocessing and partitioning.

Metric	Standard classifier	Calibrated classifier
Accuracy	97.67%	97.67%
Scam precision	88.81%	91.94%
Scam recall	92.97%	89.06%
Scam F1-score	90.84%	90.48%
False-positive rate	1.66%	1.11%
ROC-AUC	0.9951	0.9951
Scam PR-AUC	0.9718	0.9718
Brier score	0.0341	0.0171

Calibration improved the Brier score from 0.0341 to 0.0171, indicating better probabilistic predictions under this metric. At the default classification threshold, precision increased and the false-positive rate decreased, while scam recall also decreased. The unchanged ROC-AUC and PR-AUC are consistent with calibration changing the probability scale without materially changing the ranking of messages.

5.4.2 Coverage and Error Trade-off

The calibrated classifier was evaluated at eleven confidence thresholds. Selective accuracy in the table is calculated as $1 - selectiveerrorrate$. It describes only the messages classified automatically, not all test messages.

Confidence threshold	Coverage	Selective accuracy	Scam recall	Review workload
0.50	100.00%	97.67%	89.06%	0.00%
0.55	99.71%	97.86%	89.06%	0.29%
0.60	99.32%	98.15%	89.06%	0.68%
0.65	98.74%	98.33%	88.28%	1.26%
0.70	97.77%	98.61%	85.16%	2.23%
0.75	97.19%	98.70%	83.59%	2.81%
0.80	96.71%	98.80%	80.47%	3.29%
0.85	95.74%	98.99%	78.91%	4.26%
0.90	94.19%	99.38%	75.00%	5.81%
0.95	90.31%	99.79%	64.84%	9.69%
0.99	71.22%	100.00%	48.44%	28.78%

Source: reported Experiment 5 test-set results. Percentages are rounded to two decimal places. Selective accuracy is derived from the reported selective error rates.

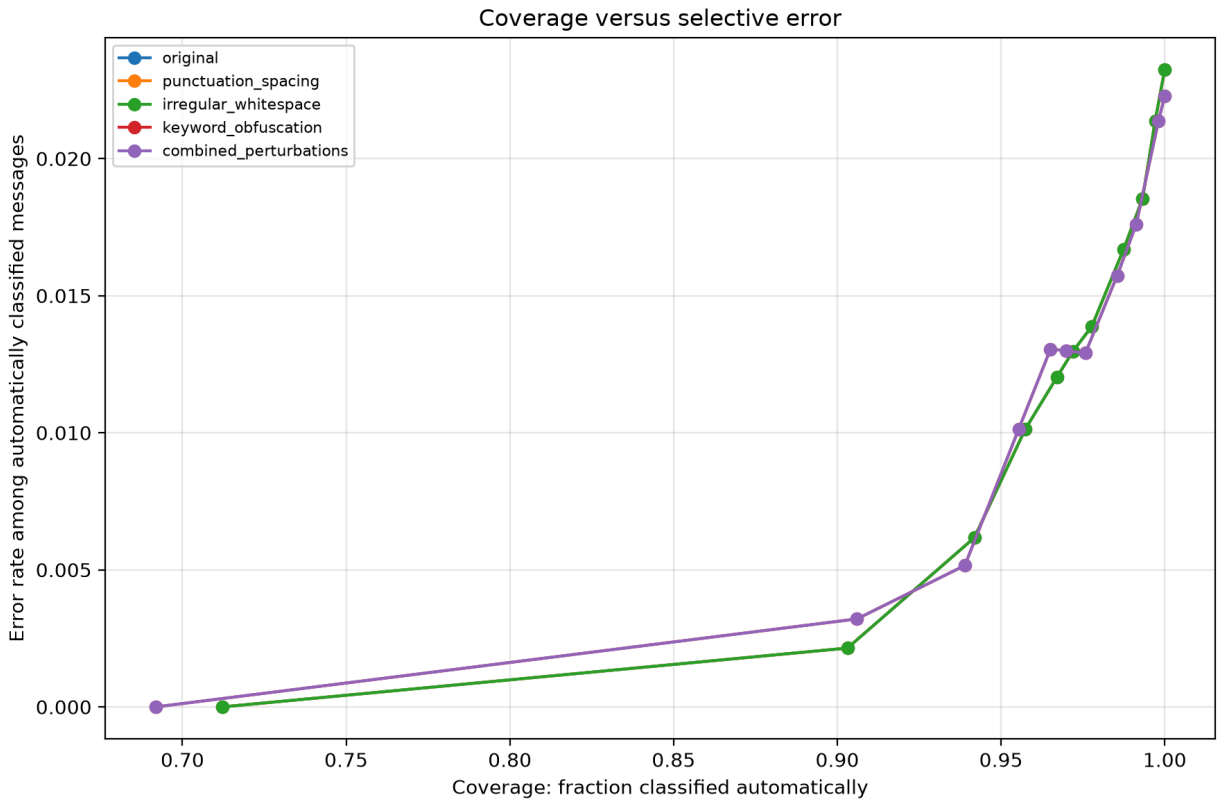
The results show that increasing the confidence threshold consistently reduced the observed selective error rate while increasing the number of messages requiring review. At a threshold of

0.90, coverage was 94.19%, selective accuracy was 99.38%, and review workload was 5.81%. However, scam recall fell to 75.00%.

At a threshold of 0.99, the reported selective error rate was zero among automatically classified messages, but scam recall fell to 48.44%, while 28.78% of messages required review. Thus, zero observed errors among automatic decisions did not imply that the system detected most scams.

These findings demonstrate that the operating threshold must account for both classification reliability and the cost of missed scams. A threshold that minimizes errors among automatically classified messages may be unsuitable if missing scams is the primary concern.

5.4.3 Adversarial Perturbation Evaluation under Abstention



The calibrated classifier was tested against punctuation spacing, irregular whitespace, keyword obfuscation, and combined perturbations.

Condition	Accuracy	Scam recall	False-positive rate	Brier score
Original	97.67%	89.06%	1.11%	0.0171
Punctuation spacing	97.67%	89.06%	1.11%	0.0171
Irregular whitespace	97.67%	89.06%	1.11%	0.0171
Keyword obfuscation	97.77%	88.28%	0.88%	0.0171
Combined perturbations	97.77%	88.28%	0.88%	0.0171

Punctuation-spacing and irregular-whitespace transformations produced no measurable change in the reported metrics. Keyword obfuscation caused a small reduction in scam recall, from 89.06% to 88.28%, while accuracy increased slightly. The combined transformation produced the same reported metrics as keyword obfuscation.

These results suggest that the tested transformations had limited effects on this dataset and model. They do not establish robustness against more sophisticated adversarial messages, previously unseen scam strategies, or distribution shifts. The perturbations were synthetic, and their effects should not be generalized beyond the transformations tested.

6.4 Operational Implications of Abstention

The experiment indicates that abstention can provide an additional layer of control in scam-detection systems by preventing low-confidence predictions from automatically triggering consequential actions. Instead of forcing every message into a binary category, a system can refer uncertain messages for human assessment.

The appropriate threshold depends on the relative costs of false positives, missed scams, and manual review. For example, a threshold of 0.90 in this experiment reduced the observed error rate among automatic decisions to 0.62%, while referring 5.81% of messages for review. However, scam recall fell to 75.00%, demonstrating that this operating point left a substantial proportion of actual scams without an automatic positive classification.

A practical deployment could therefore use abstention when automated decisions would lead to blocking a sender, warning a user, or taking another consequential action. Messages requiring review could be prioritized according to confidence, estimated harm, and available review capacity. The system should not treat an abstained message as safe; it should mark the message as unresolved until further assessment.

Several limitations remain. First, the evaluation uses a single SMS spam dataset, so results may not transfer to modern phishing messages, multilingual scams, or other communication channels. Second, probability calibration does not guarantee reliable confidence estimates under distribution shift. Third, the adversarial tests cover only a small number of synthetic text transformations. Finally, the results do not measure real human-review accuracy, review latency, or operational costs.

3.9 Deployment Efficiency Benchmark Protocol

Dataset and evaluation protocol. After removing duplicate messages and excluding messages with conflicting labels, the dataset contained 5,158 unique messages: 4,516 legitimate messages and 642 scam messages. A stratified 80:20 train-test split was generated using random seed 42, producing 4,126 training messages and 1,032 test messages. The same split was used for all four classifiers. Each model incorporated a TF-IDF text vectorizer with unigram and bigram features, a minimum document frequency of two, and a maximum vocabulary size of 50,000 features.

Latency and throughput measurement. Each fitted model underwent three warm-up predictions before timing measurements. Single-message inference latency was measured using a fixed subset of up to 500 test messages over three repetitions. The 50th percentile (p50), 95th percentile (p95), mean latency, and standard deviation were calculated from the recorded individual prediction times. Throughput was measured separately using batches of 32 messages. The serialized size was obtained from the saved, fitted joblib artifact, including the text vectorizer and classifier.

Adversarial evaluation. The models were additionally evaluated on a deterministically transformed version of the same test messages. Transformations included case changes, added whitespace, and selected character substitutions in common terms. This evaluation examined sensitivity to the specified text perturbations. Since semantic equivalence was not independently verified for every transformed message, these results are interpreted as performance under controlled perturbations rather than a definitive measure of adversarial robustness.

5.5 Deployment Efficiency Results

Table 5 presents the measured detection performance and deployment efficiency of the four classifiers.

Model	Clean Macro-F1	Clean scam recall	Perturbed Macro-F1	Perturbed scam recall	p50 (ms)	p95 (ms)	Model size (MB)	Throughput (msg/s)
Linear SVM	0.9643	0.9375	0.9596	0.9219	0.2810	0.3504	0.4406	46,036.70
SGD Classifier	0.9625	0.9453	0.9625	0.9453	0.2818	0.3739	0.4410	49,445.71
Logistic Regression	0.9502	0.9453	0.9479	0.9375	0.2813	0.3423	0.4407	45,302.17
Multinomial Naive Bayes	0.9391	0.8047	0.9337	0.7891	0.3010	0.3859	0.6712	47,539.81

Table 5. Measured detection performance and deployment efficiency. Latency is reported in milliseconds per individual prediction; throughput is reported in messages per second under batched inference.

Linear SVM achieved the highest clean-test Macro-F1 of 0.9643, with a scam recall of 93.75%. Its p50 inference latency was 0.2810 ms, and its serialized model size was 0.4406 MB. Under the specified text perturbations, its Macro-F1 decreased to 0.9596 and scam recall decreased to 92.19%.

The SGD Classifier achieved a clean-test Macro-F1 of 0.9625 and scam recall of 94.53%. It also achieved the highest measured throughput at approximately 49,446 messages per second. Its perturbed-input Macro-F1 and scam recall remained at 0.9625 and 94.53%, respectively. These measurements indicate that SGD offered a favorable combination of detection performance, throughput, and compact serialized size in this experiment.

Logistic Regression had the lowest measured p95 latency at 0.3423 ms, marginally below Linear SVM at 0.3504 ms. However, its clean-test Macro-F1 was 0.9502, below the values achieved by Linear SVM and SGD. Multinomial Naive Bayes had the largest serialized model size, at 0.6712 MB, and the lowest scam recall among the evaluated models, at 80.47% on clean inputs.

The observed latency standard deviations were 0.0743 ms for Logistic Regression, 0.0749 ms for Multinomial Naive Bayes, 0.0765 ms for Linear SVM, and 0.1056 ms for SGD. These values describe variability across individual timing observations pooled from the three measurement

repetitions. They should not be interpreted as standard deviations across independent benchmark sessions.

The results are summarized visually in the performance-versus-latency plot. Linear SVM provides the highest clean-test Macro-F1, whereas SGD provides the highest measured throughput and strongest scam recall under the evaluated perturbations. Consequently, the model with the highest detection score is not necessarily the model with the highest throughput.

Peak model-specific memory consumption was not established in this experiment. The recorded memory values were process-wide RSS snapshots and cannot isolate the memory attributable to an individual model. Therefore, conclusions about peak RAM requirements remain outside the scope of the current measurements.

6.5 Deployment Efficiency Trade-offs

The deployment results indicate that Linear SVM and SGD Classifier are promising candidates for resource-constrained scam detection systems. Both achieved clean-test Macro-F1 scores above 0.96 and serialized model sizes of approximately 0.44 MB. Their measured single-message median latency was approximately 0.28 ms on the experimental computer.

Linear SVM is the preferred candidate when maximizing clean-test Macro-F1 is the primary objective. SGD Classifier is the preferred candidate when throughput and scam recall under the tested perturbations are prioritized. In this experiment, SGD processed approximately 49,446 messages per second in the batch-throughput measurement, compared with approximately 46,037 messages per second for Linear SVM. These figures represent measured throughput in the specified environment and should not be assumed to transfer directly to other hardware or deployment settings.

Logistic Regression offers a competitive alternative, with a measured p95 latency of 0.3423 ms and a scam recall of 94.53% on clean inputs. Multinomial Naive Bayes may still be useful in settings where its behavior is preferred, but its lower scam recall in this evaluation makes it less attractive when missed scams are a major concern.

The small serialized artifacts and low measured CPU inference latencies suggest that these classifiers merit further evaluation for local deployment. However, model-file size alone does not determine actual deployment memory requirements. The present experiment did not measure isolated peak memory, cold-start cost, battery consumption, or inference performance on a representative mobile device. A mobile-deployment claim would therefore require additional measurements on the target device.

Overall, the experiment demonstrates that deployment decisions should consider detection performance, latency, throughput, model size, and robustness together rather than relying on accuracy alone. Linear SVM and SGD Classifier represent the strongest candidates under the measured conditions, with the final choice depending on whether clean-test Macro-F1 or throughput and scam recall are given greater priority.

3.10 Explainability Stability Protocol

This experiment investigated whether the explanations produced by a text-based scam detector remain stable under input transformations. Linear SVM was selected as the primary model because it achieved the highest clean-test Macro-F1 of 0.9643 in Experiment 6.

A sample of 100 correctly classified messages was selected from the existing test predictions, comprising 50 legitimate messages and 50 scam messages. The experiment evaluated two formatting transformations and one candidate paraphrase transformation. Formatting transformations involved changing character case, introducing additional whitespace, and modifying outer whitespace. The candidate paraphrase transformation applied a predefined lexical replacement when a matching phrase was present.

Feature attribution method. Word-level attribution was estimated using an occlusion-based perturbation method. For each message, individual words were removed from the input, and the change in the model's decision margin was measured. The attribution of a word was calculated as the original predicted-class score minus the score obtained after removing that word. Positive values indicate that removing the word reduces support for the original predicted class, whereas negative values indicate that its removal increases that support.

The analysis considered up to 24 unique words per message and ranked the features by the absolute magnitude of their estimated attribution. The ten highest-ranked features were used for the primary explanation-stability comparison. This method provides a local sensitivity-based explanation and does not establish causal importance.

Stability metrics. Prediction consistency was defined as the proportion of transformed messages receiving the same predicted label as their original versions. Explanation stability was evaluated using Jaccard similarity between the sets of top-ranked features. Top-feature overlap measured the proportion of shared features relative to the smaller feature set, while rank-weighted similarity assessed whether shared features retained similar relative rankings. Absolute changes in the model's decision score were also recorded.

Classification accuracy, Macro-F1, scam precision, and scam recall were calculated for the selected subset and its transformed versions. Because the original messages were selected only after correct classification, these classification metrics were interpreted conditionally rather than as estimates of full-test-set performance.

The experiment was conducted using a single model. Consequently, the results characterize the stability of Linear SVM explanations under the evaluated transformations and do not establish generalization across different classifier architectures.

5.6 Explanation Stability Results

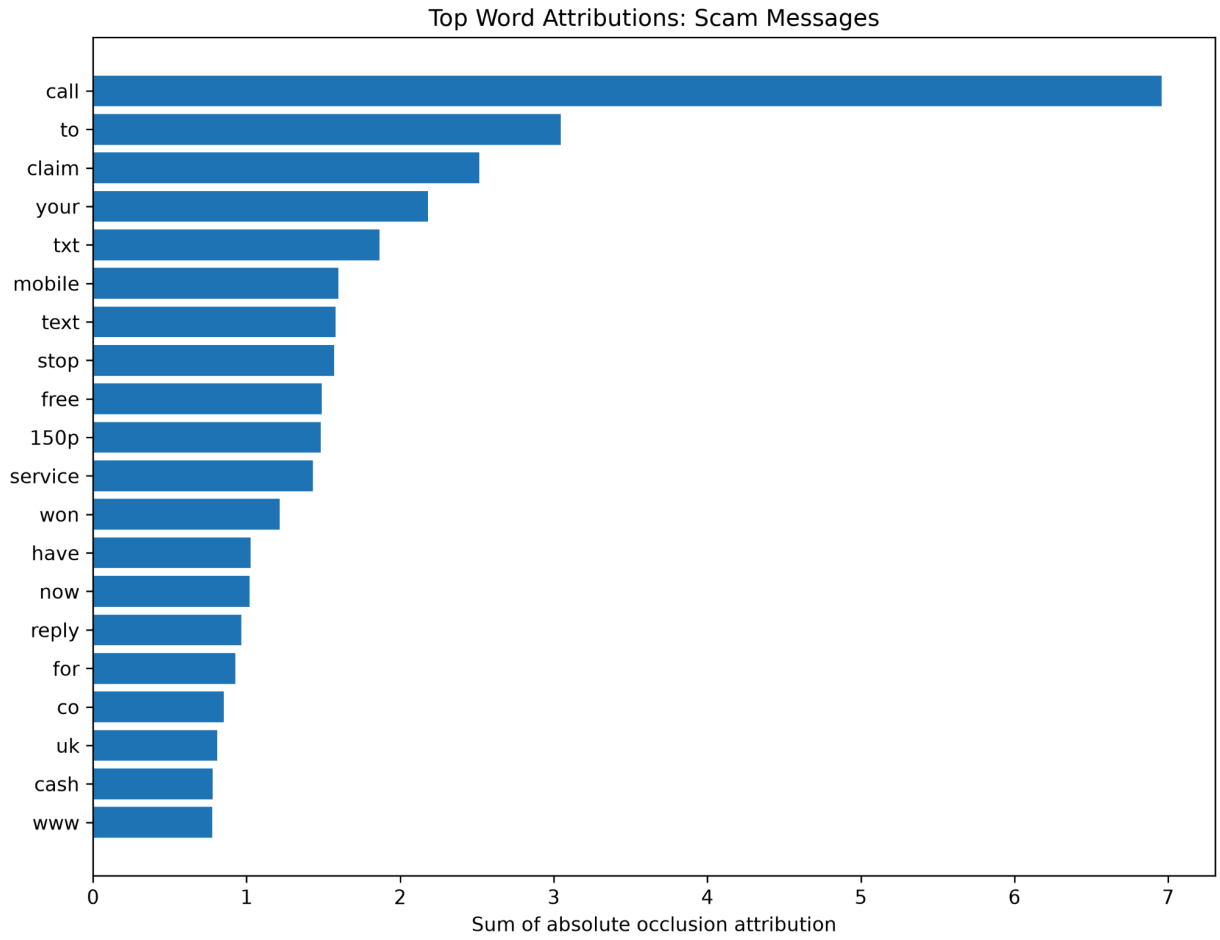
The explanation-stability experiment evaluated 100 correctly classified messages, evenly divided between legitimate and scam messages. Each message was subjected to three transformation conditions, yielding 300 message-transformation comparisons.

Prediction consistency. Linear SVM maintained 100% prediction consistency under all three evaluated conditions. No prediction changes were observed following case and whitespace modifications, outer-whitespace modifications, or the candidate paraphrase condition. The classification accuracy of the selected subset remained at 100% in each condition. These results demonstrate stability on the selected examples rather than perfect classification performance across the complete test dataset.

Explanation stability. Both formatting transformations achieved a mean top-feature Jaccard similarity of 1.000, a mean top-feature overlap of 1.000, and a mean rank-weighted similarity of 1.000. Their mean absolute decision-score changes were zero. The results indicate that these formatting modifications did not change the model's predictions, measured decision scores, or selected top-feature sets in the evaluated sample.

For the candidate paraphrase condition, mean Jaccard similarity was 0.9897, mean top-feature overlap was 0.9940, and mean rank-weighted similarity was 0.9853. The mean absolute decision-score change was 0.0060. These measurements suggest high explanation similarity for the candidate paraphrase condition, although the strength of this conclusion is limited by the transformation coverage.

Specifically, 96% of the candidate paraphrase variants were identical to their original messages because no predefined replacement rule matched. Therefore, only four of the 100 messages were actually modified by the candidate paraphrase procedure. The high similarity scores cannot be interpreted as strong evidence of paraphrase robustness.



Overall, the experiment found no prediction changes among the evaluated examples and high stability of the measured top-feature explanations. The formatting results provide evidence of invariance to the tested formatting modifications. A broader evaluation involving verified meaning-preserving paraphrases and additional classifier types is required to determine whether this stability generalizes beyond the current sample.

Results table

Transformation	Prediction consistency	Mean Jaccard	Feature overlap	Rank-weighted similarity	Mean absolute score change
Case and whitespace	100%	1.0000	1.0000	1.0000	0.0000

Transformation	Prediction consistency	Mean Jaccard	Feature overlap	Rank-weighted similarity	Mean absolute score change
Outer whitespace	100%	1.0000	1.0000	1.0000	0.0000
Candidate paraphrase	100%	0.9897	0.9940	0.9853	0.0060

Table 6. Explanation stability for Linear SVM. Candidate paraphrase results are based on only four messages that were actually modified.

6.6 Interpretability and Reliability

Interpretability and prediction reliability represent related but distinct properties of a machine-learning system. Interpretability concerns the information available to understand which input features influence a prediction. Reliability concerns whether the model's behavior remains consistent and appropriate under relevant changes in the input.

The results of Experiment 7 illustrate this distinction. Linear SVM produced word-level attribution scores that were highly consistent under the tested formatting transformations. The model's decision scores and top-ranked features remained unchanged, suggesting that the classifier was insensitive to the evaluated case and whitespace modifications.

However, stable explanations do not necessarily imply correct reasoning. A model may repeatedly rely on misleading correlations, dataset-specific vocabulary, or other features that happen to remain present after a transformation. Likewise, a stable prediction does not establish that the model correctly understands the intent of a message. Feature-attribution methods describe model behavior but cannot, by themselves, verify the semantic validity of a prediction.

The candidate paraphrase experiment provides a further limitation. Although its measured explanation similarity was high, most candidate variants were unchanged because the predefined replacement rules did not match their messages. The observed results therefore provide insufficient evidence that the model's explanations remain stable under substantial semantic-preserving changes.

The evaluation also considered only correctly classified original messages. It did not measure explanation behavior for initially incorrect predictions or determine whether the attribution method reliably identifies the features responsible for such errors. In addition, only Linear SVM

was evaluated, so the results cannot establish whether the observed stability is specific to this classifier or shared by other models.

Future evaluation should include a larger and more diverse set of manually verified paraphrases, multiple independently evaluated classifiers, and both correct and incorrect original predictions. It should also investigate examples in which predictions remain unchanged but the most influential features change substantially, as well as examples in which prediction changes coincide with shifts in feature attribution.

In conclusion, the experiment provides preliminary evidence that Linear SVM's predictions and measured explanations are stable under the specific formatting transformations tested. It does not establish semantic robustness, causal faithfulness of explanations, or cross-model generalization. These distinctions are important when interpreting explanations in security applications, where apparently understandable predictions must also be evaluated for reliability under realistic variations in message content.

7. Limitations

7.1 Source and Sampling Bias

The benchmark relies on the UCI SMS Spam Collection and reflects the characteristics of its source data, collection period, message formats, and labeling conventions. Its distribution may not represent contemporary scams, messaging applications, email phishing, multilingual fraud, or other communication channels. Results obtained from this dataset should not be assumed to generalize to these settings without additional evaluation.

7.2 Missing Metadata

The dataset does not provide campaign identifiers or sufficient sender-level metadata for campaign-grouped splitting. The experiment therefore used stratified random splitting. Although no identical normalized messages crossed the resulting splits, similar templates or related campaigns may still appear in both training and evaluation data. This may lead to optimistic performance estimates if models learn recurring templates rather than generalizable scam indicators.

7.3 Label Uncertainty

The source distinguishes ham from spam, but these categories are not equivalent to verified legitimate communication and confirmed fraud. Some spam messages may be promotional or unsolicited without being malicious, while some fraudulent messages may be labeled as legitimate. The experiment preserves the original labels and does not claim independent verification of each message's intent.

7.4 Duplicate-Detection Limitations

Exact duplicate removal was performed using normalized text and label. The near-duplicate procedure identified 124 candidate pairs based on character similarity, but these candidates require manual review. Character similarity can flag legitimate variations of common messages and may fail to identify semantically similar messages with substantially different wording. Thus, the current audit reduces exact duplication but does not establish complete elimination of near-duplicate or campaign-level leakage.

7.5 Scope of Generalization

The benchmark contains English SMS messages and does not establish multilingual capability or robustness across other channels. Further evaluation should include independently documented datasets, verified labels where feasible, more diverse message sources, and group-aware splitting when reliable campaign metadata is available.

8. Conclusion

In this work, we systematically evaluated text-based scam and spam classification across dataset construction, template generalization, adversarial defenses, multilingual cross-lingual transfer, selective prediction, deployment latency, and explanation stability. Our results confirm that relying on clean-data benchmark metrics or randomized evaluation splits can produce overly optimistic performance estimates due to template leakage. Furthermore, while text normalization and probability calibration effectively mitigate surface perturbations and false-positive rates, fine-tuning multilingual models with target-language translation augmentation is essential for robust cross-lingual generalization. Finally, benchmark comparisons demonstrate that lightweight models like Linear SVM and SGD Classifier provide an exceptional balance between high macro-F1 accuracy, low median latency (<0.3 ms), and high batch throughput on modern hardware. Future research should prioritize chronological dataset evaluations, verified naturally occurring multilingual benchmarks, and end-to-end human-in-the-loop operational trials.

References

Almeida, T. A., Hidalgo, J. M. G., & Yamakami, A. (2011). Contributions to the study of SMS spam filtering: New collection and results. *Proceedings of the 2011 ACM Symposium on Document Engineering*, 259–262.

Almeida, T., & Hidalgo, J. (2011). SMS Spam Collection [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5CC84

Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20(3), 273–297.

Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 4171–4186.

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1135–1144.

Wang, L., Yang, N., Huang, X., Jiao, B., Yang, L., Jiang, D., Majumder, R., & Wei, F. (2020). Textembeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533*.